

Inferring generalized parton distributions from DVCS observables

A neural posterior solution to an ill-posed inverse problem

Full-sector synthetic closure and native-model holdout tests

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31 August 2026

The target is a posterior over the full modeled GPD content

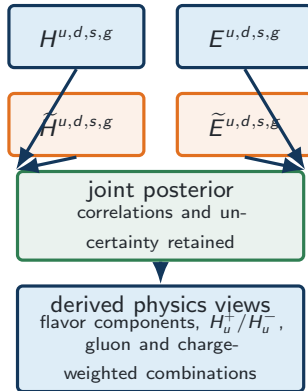
Generalized parton distributions connect longitudinal momentum and transverse spatial structure while retaining the amplitude-level information probed by exclusive scattering. The learned object is

$$q_\varphi(\theta, \eta \mid D), \quad \dim(\theta, \eta) = 80 + 2,$$

with independent modeled sectors

$$F \in \{H, E, \tilde{H}, \tilde{E}\}, \quad p \in \{u, d, s, g\}.$$

Posterior samples define the corresponding GPD curves and, after exact PARTONS reevaluation, CFFs and observables.



H_u^+ and H_u^- are representative derived components—not the unique or primary network output.

DVCS constrains convolutions, not pointwise GPD values

At each measured (x_B, t, Q^2, ϕ) , the native calculation performs

$$F(x, \xi, t, Q_0^2) \xrightarrow{\text{QCD evolution}} F(x, \xi, t, Q^2),$$
$$\mathcal{F}(\xi, t, Q^2) = \int_{-1}^1 dx \, C(x, \xi, Q^2) F(x, \xi, t, Q^2),$$

where $\mathcal{F} \in \{\mathcal{H}, \mathcal{E}, \tilde{\mathcal{H}}, \tilde{\mathcal{E}}\}$ is a complex Compton form factor (CFF).

The measured quantity is then built from

$$|T_{\text{BH}} + T_{\text{DVCS}}|^2 = |T_{\text{BH}}|^2 + 2\Re(T_{\text{BH}} T_{\text{DVCS}}^*) + |T_{\text{DVCS}}|^2.$$

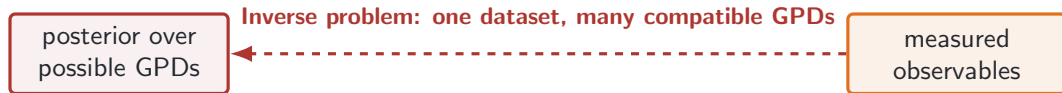
Different observables probe different CFF combinations, but none is a direct measurement of $H^u(x, \xi, t)$.

Six complementary observables

- ▶ unpolarized / beam-spin half-sum
- ▶ beam-spin difference
- ▶ beam-charge asymmetry
- ▶ beam-spin asymmetry
- ▶ longitudinal target-spin asymmetry
- ▶ longitudinal double-spin asymmetry

The observable does not uniquely determine the underlying GPD

Forward problem: well defined

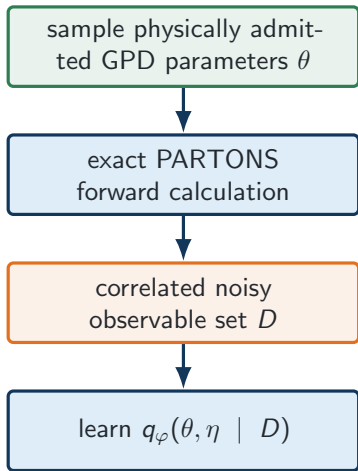


Information is compressed
CFF convolution integrates over x .

Directions are degenerate
Correlated GPD changes can leave observables nearly unchanged.

Coverage is finite
Limited kinematics and uncertainty amplify ambiguity.

We learn a posterior inverse map, not a pointwise inverse

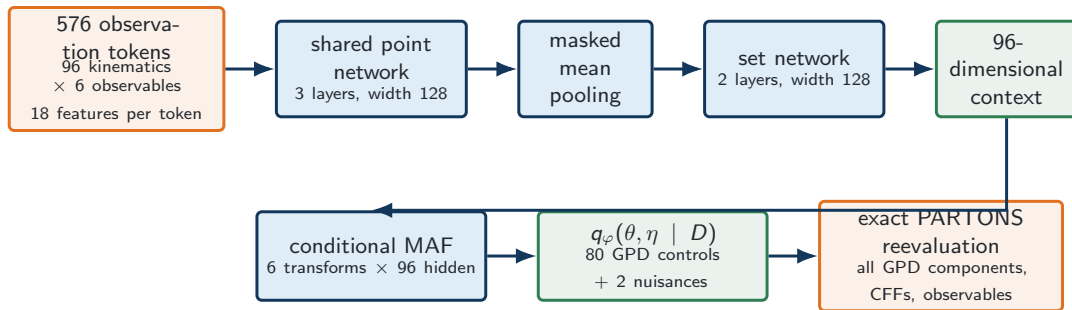


$$\mathcal{L}_{\text{NPE}} = -\mathbb{E}_{\theta, D}[\log q_\varphi(\theta, \eta \mid D)] .$$

The network learns which regions of the declared physics parameter space are compatible with an unordered set of observables. Posterior samples are then reevaluated through the exact native calculation to obtain CFF and GPD bands.

No uniqueness is imposed: weakly identified directions should remain broad.

Physics information is compressed before posterior inference



Each token contains kinematics, observable identity, normalized value, uncertainty, full-covariance whitening information, nuisance responses, and a mask. Permutation-invariant pooling makes the inference depend on the dataset, not its ordering.

The training corpus is generated from a broad physics model space

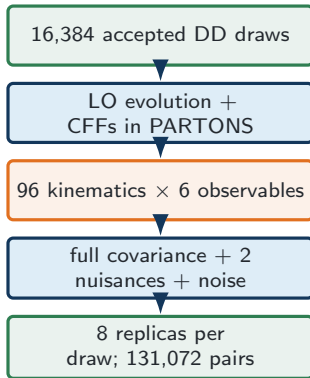
At $Q_0^2 = 1 \text{ GeV}^2$, each of $F \in \{H, E, \tilde{H}, \tilde{E}\}$ and $p \in \{u, d, s, g\}$ has an independent double-distribution block:

$$F^p = \int d\beta d\alpha \delta(x - \beta - \xi\alpha) f_{F,p}(\beta, t) \pi_{b_{F,p}}(\beta, \alpha),$$

$$f_{F,p} \propto N_{F,p} \beta^{a_{F,p}} (1 - \beta)^{c_{F,p}} e^{B_{F,p} t}.$$

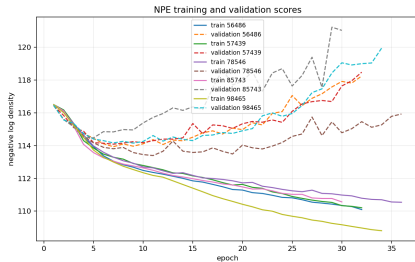
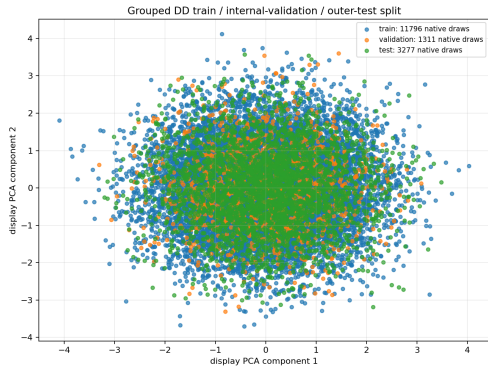
Five controls (N, a, c, b, B) for each of 16 blocks give **80 independent physics coordinates**.

Declared support includes $N \in (-4, 4)$, $a \in (0.1, 1)$, $c \in (2, 6)$, $b \in (1, 4)$, and $B \in (0, 2) \text{ GeV}^{-2}$.



The corpus spans a declared DD-induced solution space; it is broad, but not model independent.

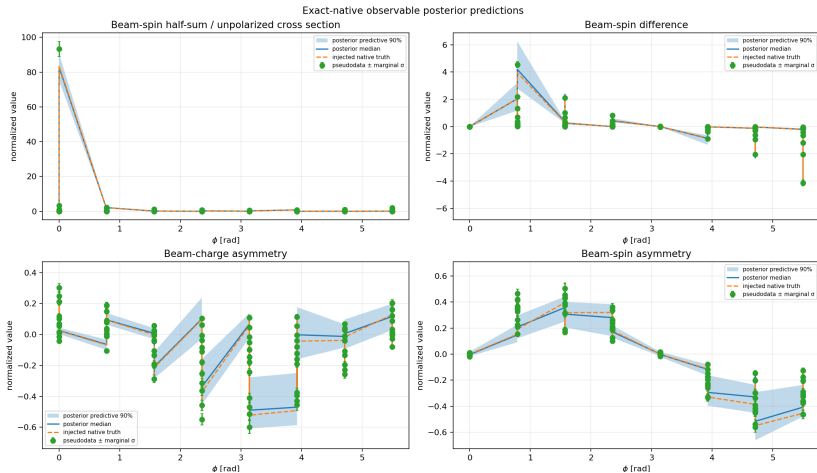
Grouped splits prevent noisy replicas from leaking across roles



- ▶ 11,796 train / 1,311 internal validation / 3,277 locked outer-test parameter groups.
- ▶ Replicas from one physics draw remain in the same role.
- ▶ Validation NLL turns upward after its minimum: later epochs overfit.
- ▶ Early stopping selects the validation minimum; the outer test is never used for training or member selection.

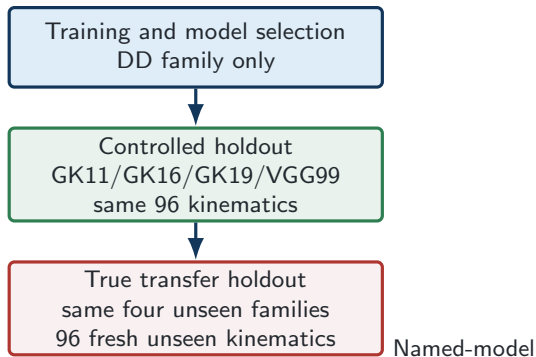
Selected ensemble: mean train NLL 112.76, mean outer-test NLL 113.91; worst test-validation gap 0.112.

Posterior samples reproduce a withheld synthetic observable set



Four representative panels are shown; the stored check includes all six observables.

A true holdout changes both the physics family and the kinematics



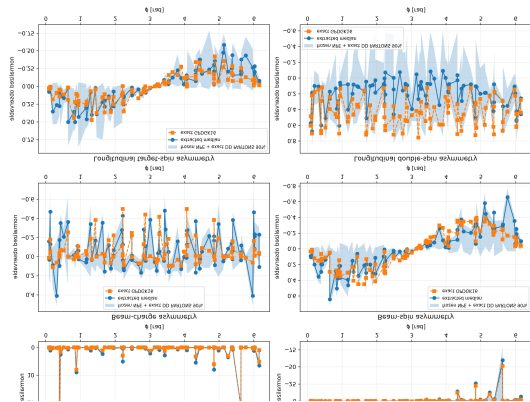
outputs were absent from corpus generation, optimization, early stopping, and ensemble selection. The frozen network is evaluated once; the model is not retuned on holdout results.

Pointwise observable truth coverage by the nominal 90% bands

model	same kinematics	fresh kinematics
GK11	0.806	0.759
GK16	0.859	0.780
GK19	0.797	0.747
VGG99	0.764	0.870

Both holdout campaigns fail the predeclared predictive-precision gate.

Fresh-design holdout exposes observable-level bias



Representative case: GK16

90% truth coverage

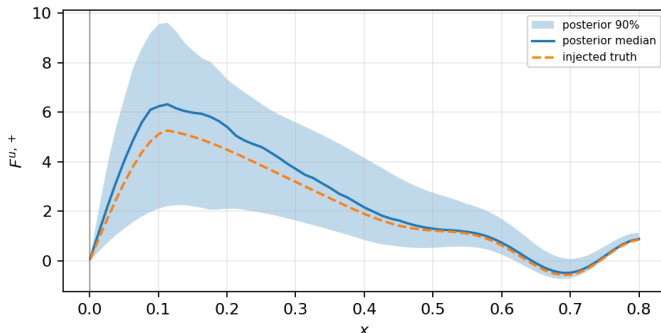
observables	0.780
CFF points	0.780
GPD points	0.965

64/64 posterior samples were valid under exact reevaluation, so the mismatch is scientific rather than an execution failure.

Broad GPD bands can cover truth while the predicted observables remain biased.

The current network has learned useful DD-family structure, but not yet a transferable inverse map.

Representative GPD diagnostic: synthetic closure for H_u^+

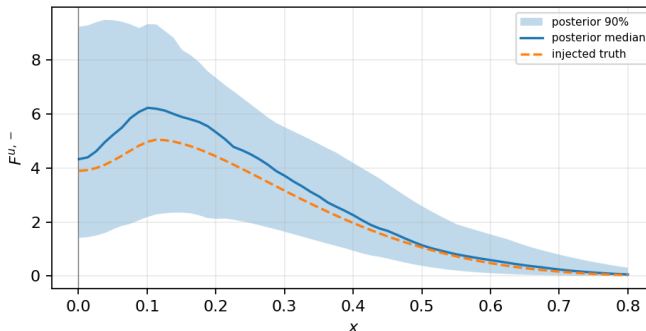


Exact PARTONS reevaluation at the declared reference kinematics
 $x_B = 0.20$, $t = -0.12 \text{ GeV}^2$, $Q^2 = 2.0 \text{ GeV}^2$.

Blue band: central 90% posterior
blue line: posterior median
orange dashed: injected truth

The median follows the injected C-even shape; uncertainty is largest at low x and honestly retains the truth.

Representative GPD diagnostic: synthetic closure for H_u^-



Exact PARTONS reevaluation at the same reference kinematics and with the same posterior samples as the H_u^+ panel.

Blue band: central 90% posterior
blue line: posterior median
orange dashed: injected truth

The C-odd/valence-like trend is reproduced, but the low- x band is too broad to claim a precise separation from H_u^+ .

What has been demonstrated—and what has not

Demonstrated

- ▶ A DeepSets-conditioned flow learns an 82-dimensional posterior inverse map within the declared DD corpus.
- ▶ Locked outer-test scores, repeated coverage, posterior predictive checks, and exact reevaluation pass the synthetic gates.
- ▶ The output spans all modeled $H, E, \tilde{H}, \tilde{E}$ flavor and gluon sectors; H_u^+ and H_u^- provide representative curve-level closure views.

Current physics limitation

- ▶ Fresh GK/VGG families and unseen kinematics expose observable-level bias: nominal 90% truth coverage is 0.747–0.870.
- ▶ High holdout GPD coverage is driven partly by broad bands; it is not evidence of precise recovery.
- ▶ The result is a synthetic closure study, not a real-data extraction. Real-data inference remains disabled.

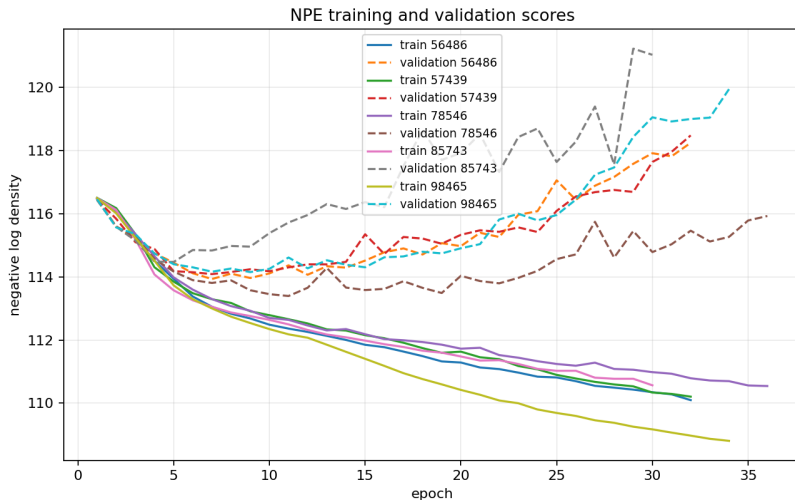
The next physics challenge is representation robustness: enlarge the training solution space before interpreting any extracted GPD component from data.

Complete validation catalogue

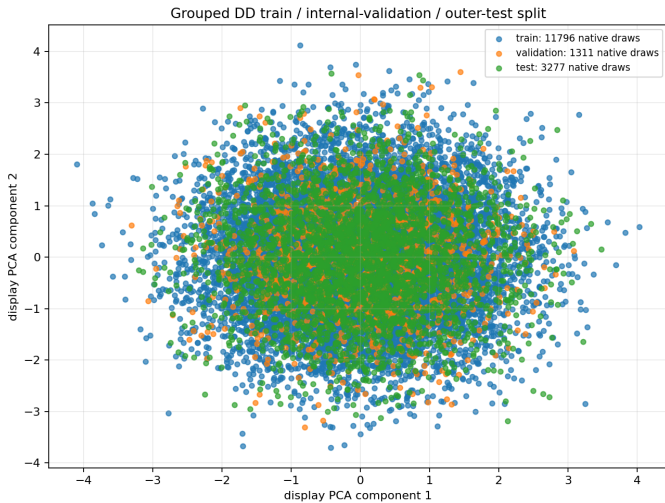
All 19 current training/validation figures and all 24 controlled- and true-holdout figures

One plot per slide preserves labels, bands, and discrepancies at projector scale. Appendix numbering is excluded from the 15-slide core talk.

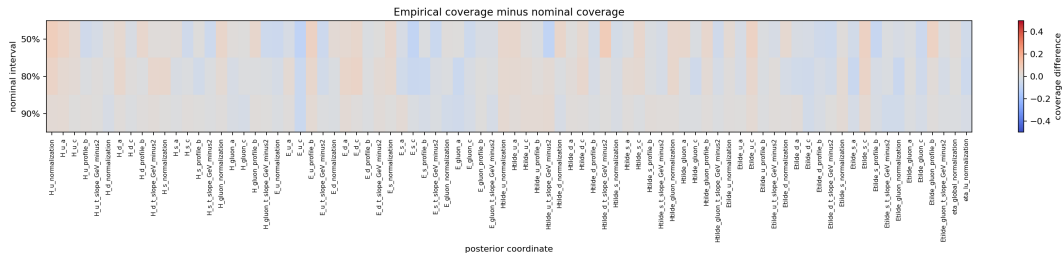
Training history: member-wise train and validation NLL



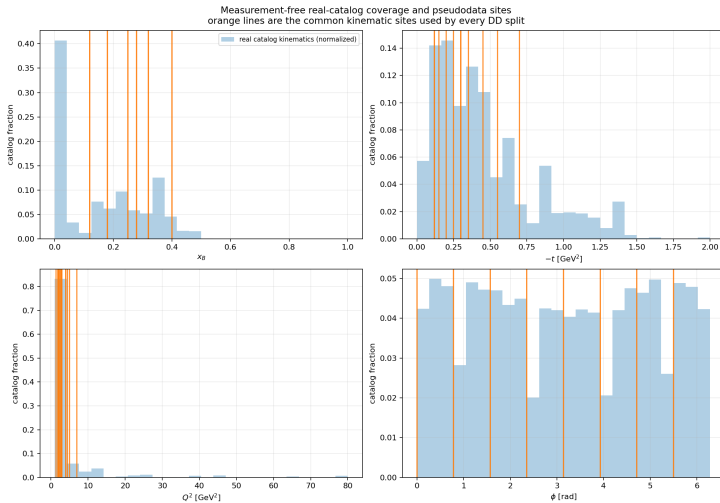
Corpus split: grouped train, validation, and outer test



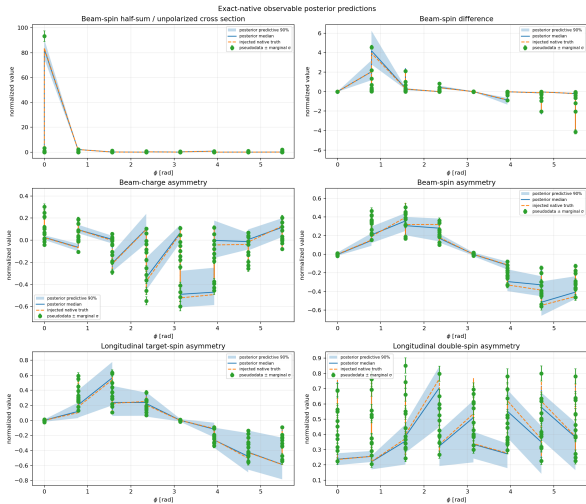
Synthetic calibration: empirical credible-interval coverage



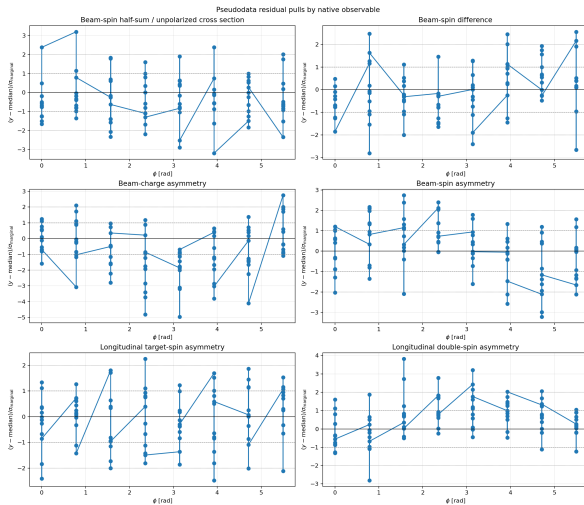
Kinematic support: real catalogue and synthetic pseudodata



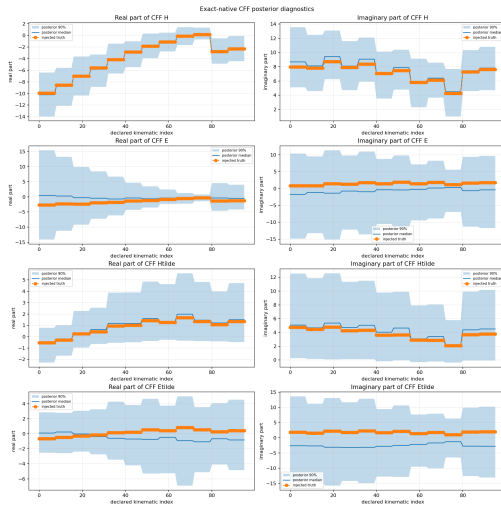
Outer-test posterior predictive distributions



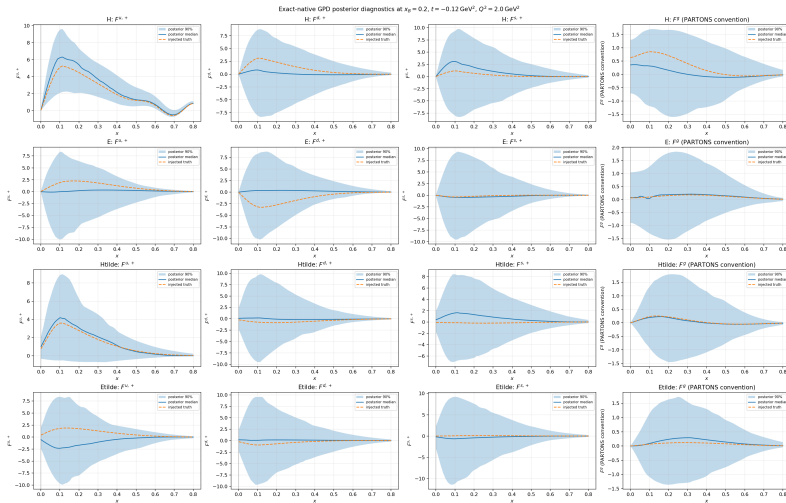
Outer-test posterior predictive pulls



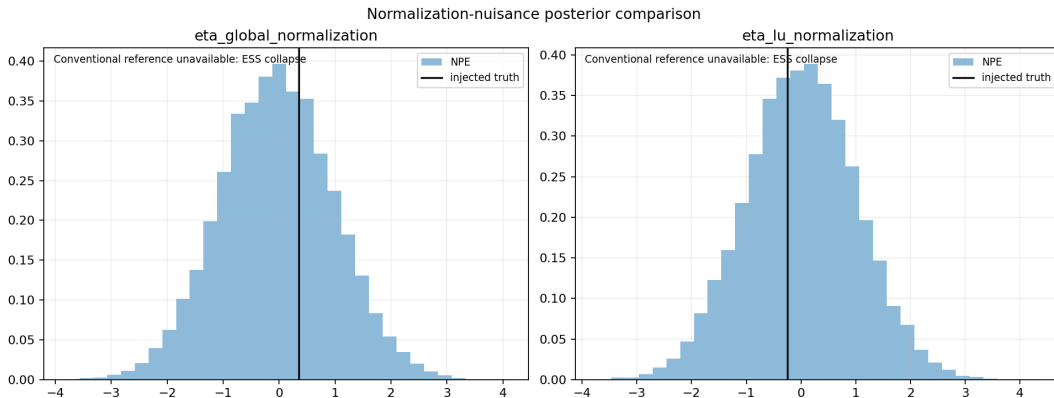
Outer-test CFF real and imaginary components



Outer-test GPD posterior predictions

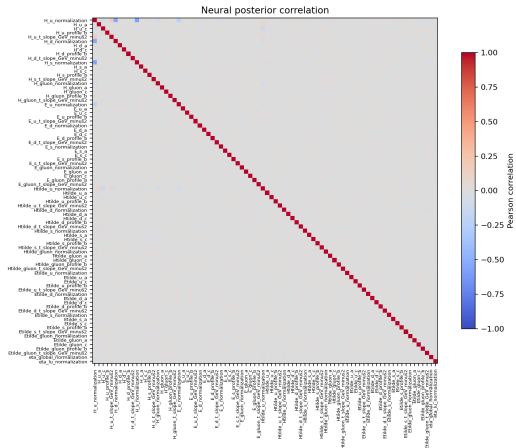


Posterior estimator comparison

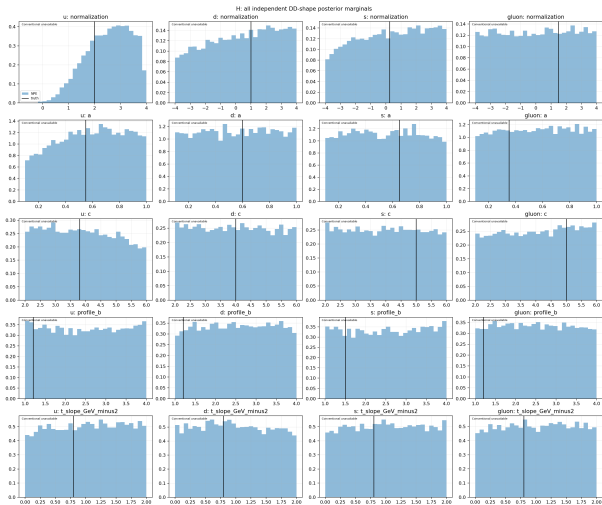


Posterior correlation structure

Conventional reference unavailable
importance-sampling ESS collapse



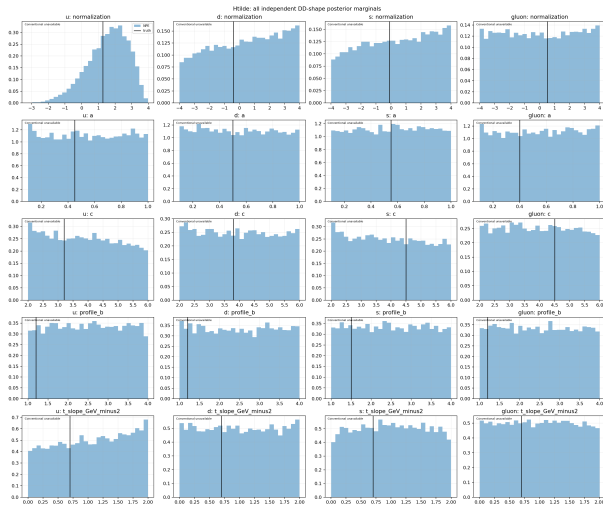
Posterior marginals: H sector



Posterior marginals: E sector



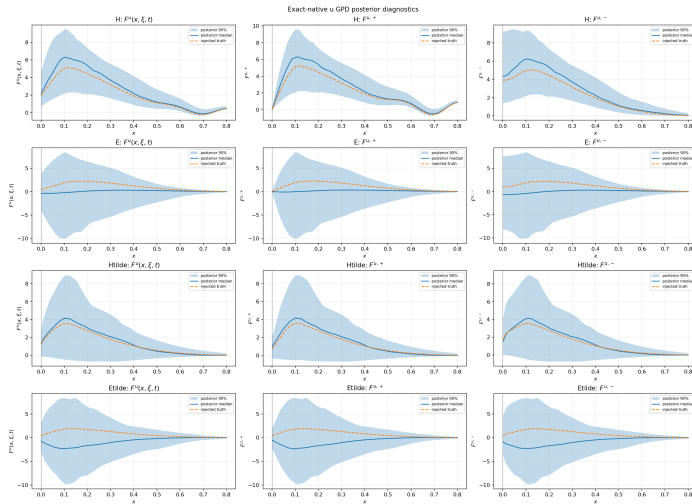
Posterior marginals: $\tilde{H}$ sector



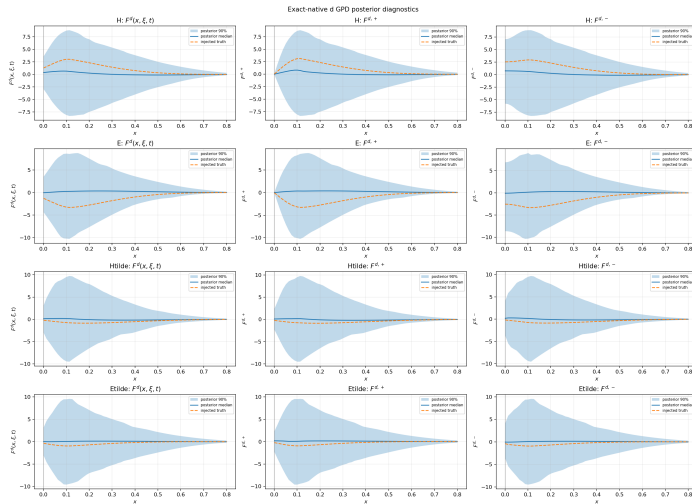
Posterior marginals: $\tilde{E}$ sector



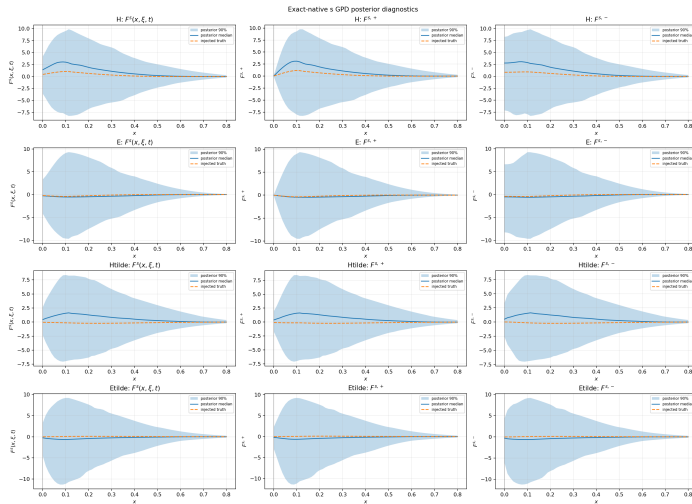
Derived GPD curves: up-quark components



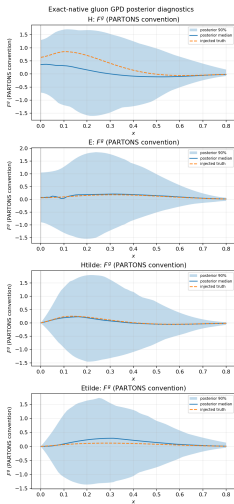
Derived GPD curves: down-quark components



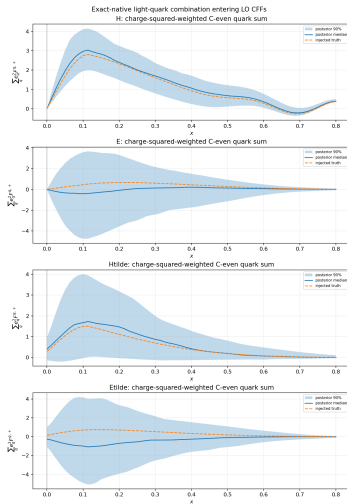
Derived GPD curves: strange-quark components



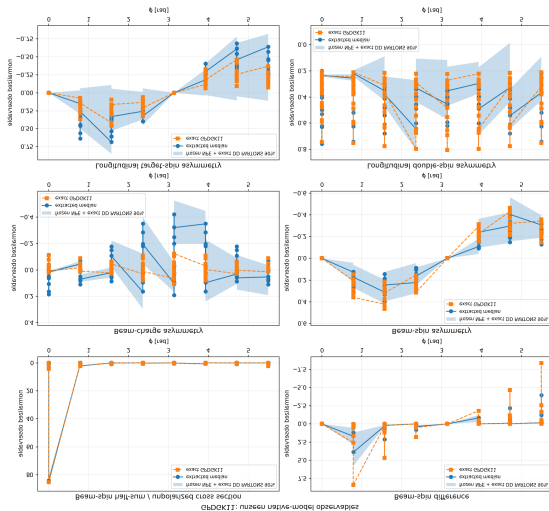
Derived GPD curves: gluon components



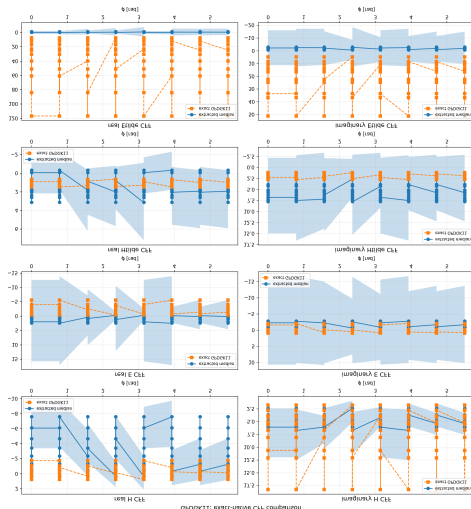
Derived GPD curve: charge-weighted C-even combination



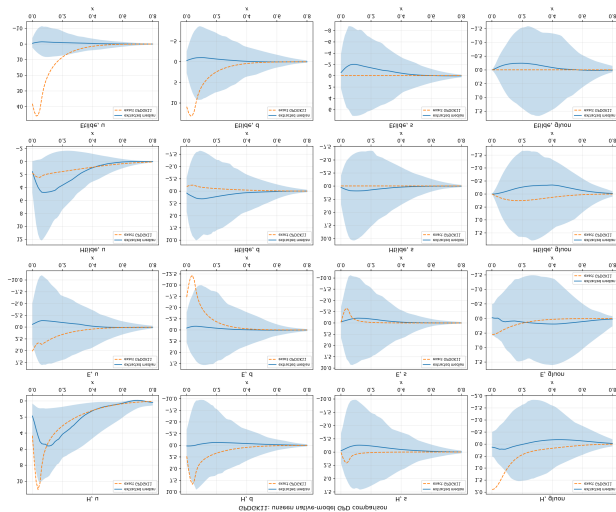
Controlled family holdout—GK11 observables



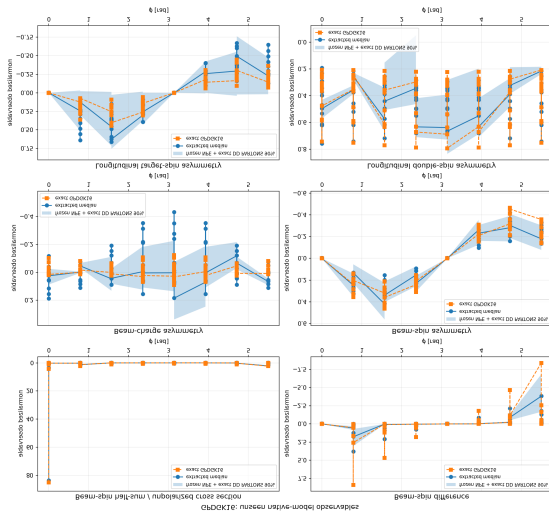
Controlled family holdout—GK11 CFFs



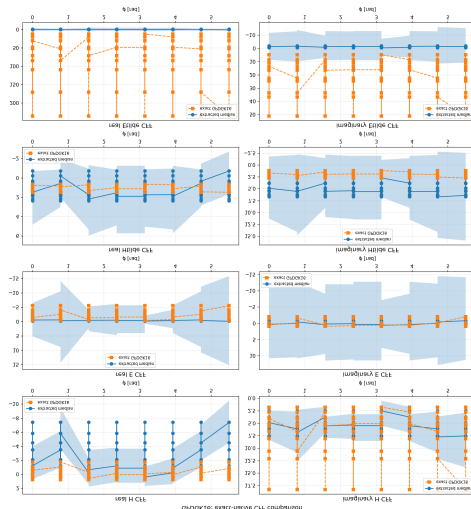
Controlled family holdout—GK11 GPDs



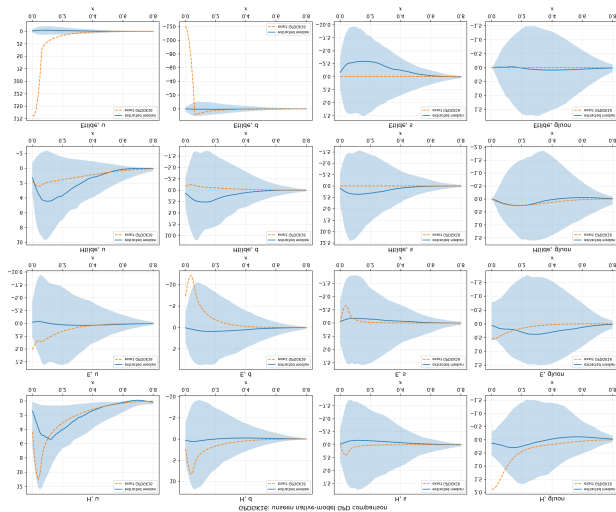
Controlled family holdout—GK16 observables



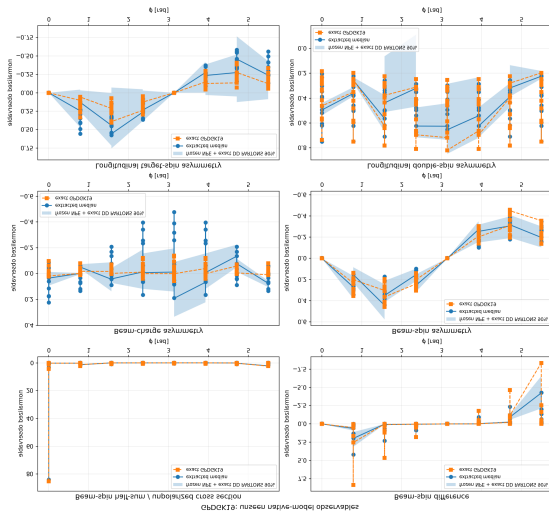
Controlled family holdout—GK16 CFFs



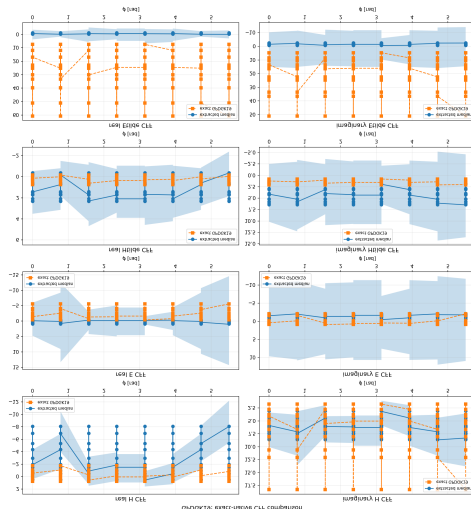
Controlled family holdout—GK16 GPDs



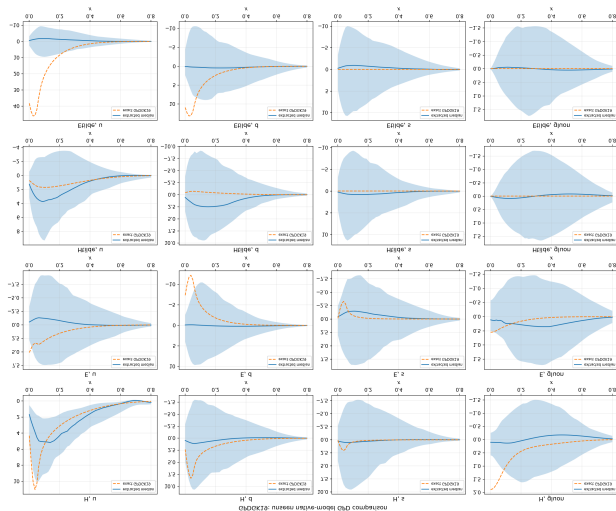
Controlled family holdout—GK19 observables



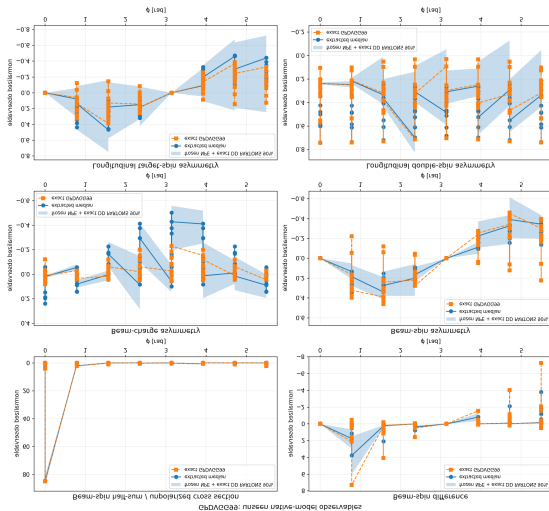
Controlled family holdout—GK19 CFFs



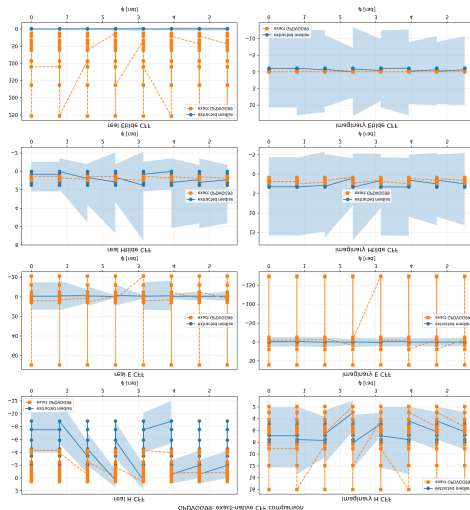
Controlled family holdout—GK19 GPDs



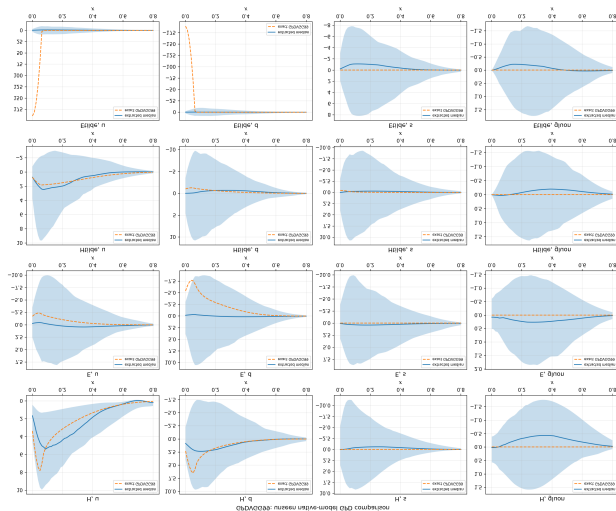
Controlled family holdout—VGG99 observables



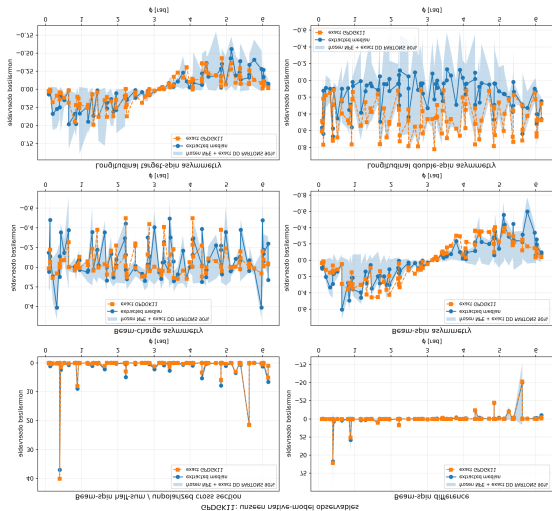
Controlled family holdout—VGG99 CFFs



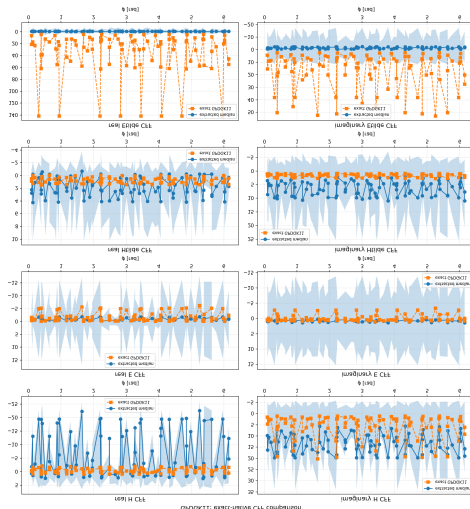
Controlled family holdout—VGG99 GPDs



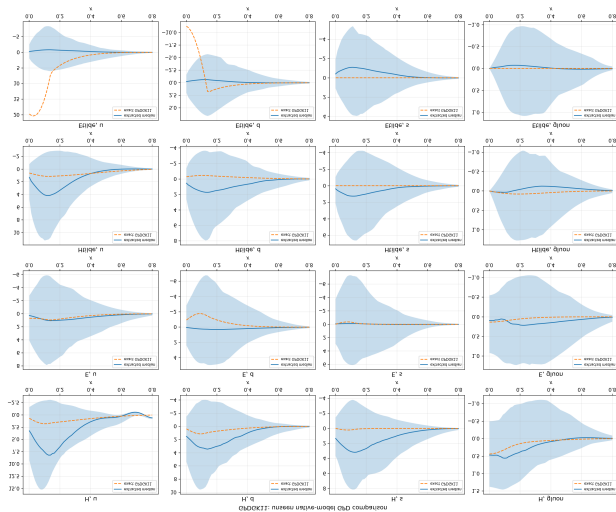
True family + kinematics holdout—GK11 observables



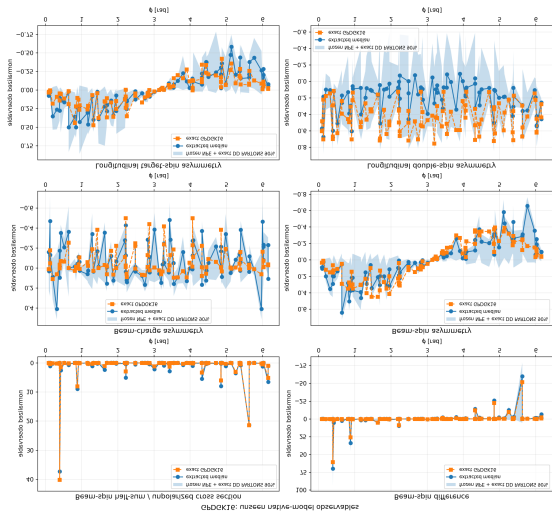
True family + kinematics holdout—GK11 CFFs



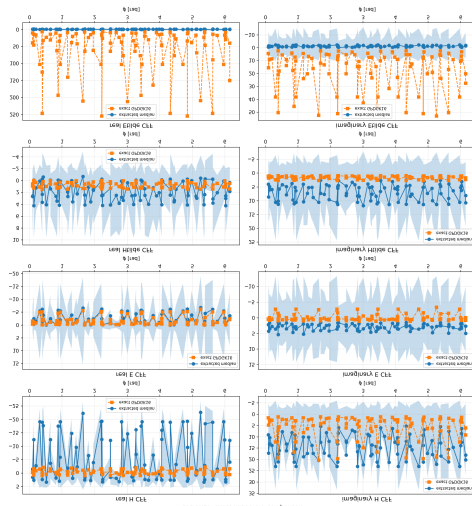
True family + kinematics holdout—GK11 GPDs



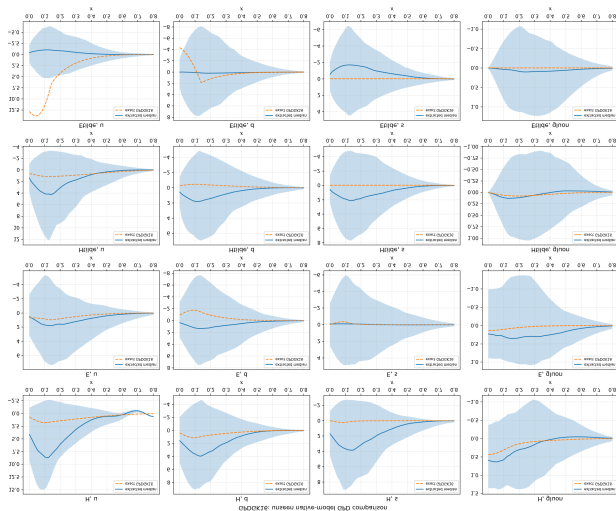
True family + kinematics holdout—GK16 observables



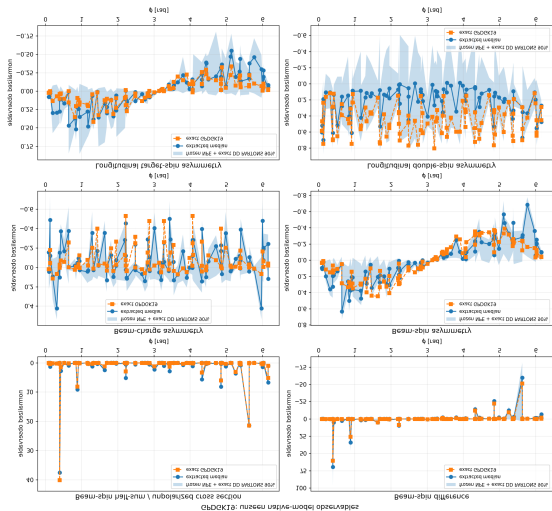
True family + kinematics holdout—GK16 CFFs



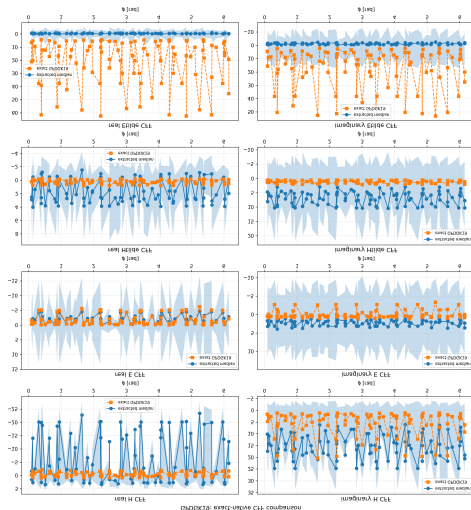
True family + kinematics holdout—GK16 GPDs



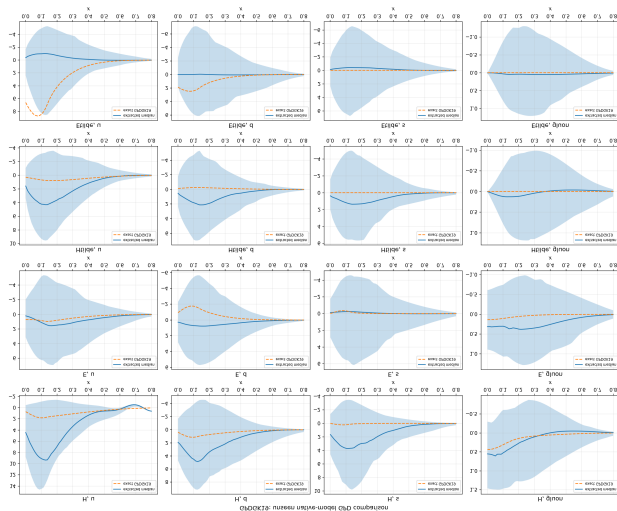
True family + kinematics holdout—GK19 observables



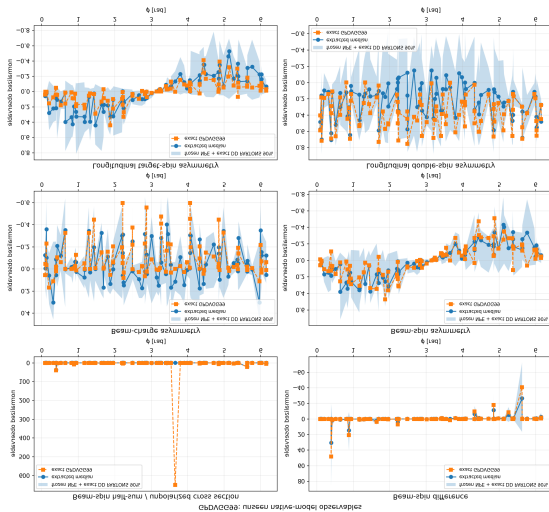
True family + kinematics holdout—GK19 CFFs



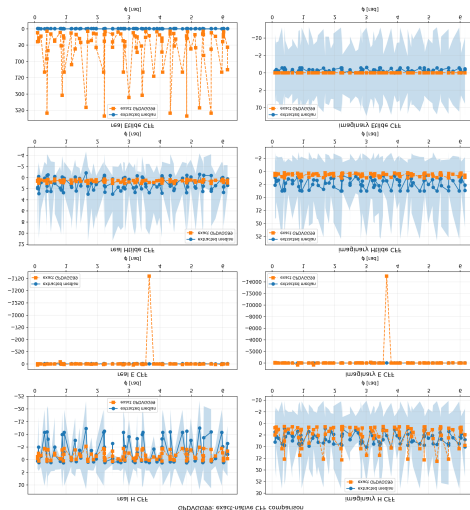
True family + kinematics holdout—GK19 GPDs



True family + kinematics holdout—VGG99 observables



True family + kinematics holdout—VGG99 CFFs



True family + kinematics holdout—VGG99 GPDs

